Distributional Pedagogical Alignment: An Evidence-Centered Framework for Knowledge Tracing

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***Abstract*—Knowledge Tracing (KT) models a learner's evolving knowledge state from historical interactions to predict future performance. Recent work on KT has advanced in three directions, among others: richer interaction representations, distributional and uncertainty-aware knowledge states, and reasoning or interpretability mechanisms. We take an evidence-centered view of this task: an interaction supports a knowledge update only after it has been aggregated and interpreted under a pedagogical lens. We treat this interpretation as an explicit intermediate stage rather than leaving it as an implicit update computation. We propose a framework organized into four representation spaces: Interaction, Distribution, Knowledge, and Prediction. At its core is the *Distributional Pedagogical Alignment* (DPA) module: interaction representations are projected into a distributional space, where learning patterns, which are structurally consistent across learners, are constructed and aligned with pedagogical principles. This yields the evidence that drives the knowledge-state update. The alignment applies rule-shaped, differentiable adjustments to distributional patterns: the functional form of each adjustment encodes a pedagogical principle, while only its magnitude is learned, so pedagogical theory constrains the form of the update rather than acting through gradient descent alone. This paper presents the framework and its rationale as a conceptual foundation for ongoing research on evidence-centered knowledge tracing.**

***Index Terms*—Knowledge Tracing, Evidence-Centered Design, Learning Pattern, Distributional Representation, Pedagogical Alignment**

I. Introduction

Knowledge Tracing (KT) models a learner's evolving knowledge state from learning interaction history to predict future outcomes [1], [2]. Recent KT models have advanced interaction representation, uncertainty modeling, and reasoning, each in complementary ways [3]–[11]. Educational measurement theory views the task as inference from evidence: a learner's proficiency is not directly observable and must be inferred from observations. A measurement model characterizes the "weight and direction" of the evidence these observations carry about proficiency [12]. Formative assessment reinforces this view by treating each interaction as evidence to be elicited, interpreted, and used to inform instructional decisions [13]. This evidence-centered view serves as the fundamental principle of the proposed framework: an interaction does not directly alter knowledge but contributes to an update only after being aggregated and interpreted through a pedagogical lens.

Concretely, the proposal introduces an explicit layer, *Distributional Pedagogical Alignment* (DPA), between the interaction representation and the mastery update, whose task is to interpret the evidence before it is used. This layer (i) projects interactions into a distributional space and organizes them into *learning patterns* that carry pedagogical meaning, (ii) aligns those patterns with pedagogical principles, and (iii) keeps the pattern structure consistent across learners. We organize the overall framework around this layer as four representation spaces with distinct roles: Interaction, Distribution, Knowledge, and Prediction (Fig. 1).

The contribution of this paper is threefold. First, a *perspective*: it frames the knowledge update as an act of interpreting evidence, grounded in pedagogical theory. Second, a *proposed mechanism*, *Distributional Pedagogical Alignment*, that gives this interpretation an explicit stage of its own, aligning learning patterns with pedagogical principles in a distributional space. Third, an *architecture* that organizes the framework into four representation spaces.

II. Related Work and Positioning

Recent KT research follows three complementary themes. **Interaction representation** enriches interaction encoding with question semantics, difficulty, and knowledge-component (KC) topology [3], [4]. **Distributional and uncertainty-aware representation** models the knowledge state, or the concepts, as distributions rather than fixed points, which gives a vocabulary for confidence and similarity that point representations lack [5]–[7]. **Reasoning and interpretability** makes inference more transparent through graph-based, probabilistic, or neural-symbolic reasoning [8]–[11].

Positioning. The framework centers on one mechanism: an explicit, differentiable alignment of learning patterns with pedagogical principles, placed between the interaction representation and the mastery update. The patterns are structured consistently across learners, so each principle applies uniformly across histories, shaping them before the mastery update is inferred. Prior KT models perform this evidence interpretation implicitly, entangle it with the mastery update, or do not cast it in pedagogical terms.

Grounded in Evidence-Centered Design (ECD) [12], this mechanism reframes the aforementioned three themes into a unified evidence interpretation stage. KC-topology-aware encoding provides the observations, a distributional coordinate enables pooling while maintaining dispersion, and a principle-guided adjustment transforms pooled patterns into evidence before, and separately from, the mastery update. Although distributional representations [6], [9], [11], pattern pooling [11], and pedagogically informed modeling [10] have each been explored independently, none introduces an explicit, pedagogically constrained alignment step between the interaction representation and the student-model update. This separation aligns with ECD principles, whereas previous approaches have left it implicit or entangled. The novelty is formulating this process as a differentiable stage, with a functional form determined by pedagogical considerations.

PLKT [11] is the closest prior work: it retrieves distribution representations from a learned, interaction-keyed lookup matrix, pools them into patterns via temporal sliding windows, and scores them by divergence from the target outcome. We instead project the history representation into a distributional space, pool by pedagogical operators, and align against pedagogical principles before inferring the mastery update. The patterns are thus simultaneously history-conditioned and pedagogically structured. They affect the mastery update rather than directly inferring the outcome. LPKT [14] is close from a different angle: it models the learning process with explicit learning and forgetting gates, but constructs neither learner-consistent patterns nor a distributional evidence stage. These are alternative design choices for individual components and form the comparison set for the planned evaluation (Section VI).

III. Problem Formulation

A learner's interaction history is a sequence $S_T = \{(q_t, r_t)\}_{t=1}^{T}$, where q_t is the question (item) attempted at step t and $r_t \in \{0, 1\}$ indicates an incorrect/correct response. The KT task is to predict $\hat{y} = P(r_{T+1} = 1 \mid q_{T+1}, S_T)$. The domain's knowledge structure is a set of knowledge components (KCs) $\mathcal{C} = \{c_1, \ldots, c_C\}$, a question-to-KC mapping, and a knowledge graph $\mathcal{G}$ encoding heterogeneous relations (e.g., prerequisite, neighboring, applying) among KCs, read by the pattern constructor of Module 2. We treat $\mathcal{G}$ as a given, validated input, and leave learning the topology itself as a promising extension [7].

The framework threads a learner's history through four representation spaces, introduced with each module in Section IV; we leave the transformation equations to our future implementation paper. We use *evidence* for the distributional representation in the alignment layer *after* it has been pooled into patterns and pedagogically adjusted: a post-alignment object. A *learning pattern* is the output of a fixed aggregation operator, defined by the aggregation rather than by the number of interactions it draws on.

IV. Proposed Framework

A. Design Rationale: From Principle to Architecture

Evidence-centered design separates three kinds of models [12]. A "student model" carries the proficiency estimate. An "evidence model" turns observations into evidence. A "task model" specifies the tasks that produce those observations, together with their features and difficulty. The proposed spaces follow these models. *Interaction* holds the observation, which is the learner's response to a given item. *Distribution* is the evidence model, where observations are pooled into patterns and aligned into evidence. *Knowledge* is the student model, which the gated update advances. The task model corresponds to the given task structure: the question and concept difficulty, the knowledge-graph topology, and the question-to-KC mapping. Knowledge tracing receives this structure rather than designing it, so it enters the framework as a fixed input. *Prediction* is an added space that maps the student model to the next-step outcome.

The spaces form a *chain of preconditions*, in which each stage enables the next. A distributional representation preserves a dispersion that a point vector discards, so pooling retains how concentrated the evidence is. The constructed patterns are structurally consistent across learners, which makes pedagogical alignment well-posed: different histories yield different distributions, but a fixed operator applied to any history carries the same pedagogical semantics. We separate the two notions this relies on: a shared *coordinate*, which is the space of distributional descriptors (mean and variance), and shared *semantics*, which is the fixed operator. A shared principle can then apply to every history because it is stated in this shared coordinate. Finally, the explicit knowledge state lets the update attribute evidence to specific KCs.

B. Overview

Fig. 1 illustrates the framework, which comprises four sequential modules, one per space, and each realizing that space's role in the preconditional chain stated above. Module 1 learns the interaction representation. Module 2 turns that representation into a distributional one, pools it into learning patterns, aligns them with pedagogical principles, and reads out the structured evidence. Module 3 updates the explicit knowledge state from that evidence while learning a pattern-to-KC contribution. Module 4 aggregates the updated state with the target question to produce the prediction and a KC-to-prediction contribution.

C. Module 1: Interaction Representation Learning

The first module learns a contextual representation for each interaction from two input groups with distinct learning dynamics. The *interaction context* gathers question semantics, the student response, and question difficulty. The *knowledge context* combines a localized mastery representation with concept difficulty. The localized mastery is obtained by reading

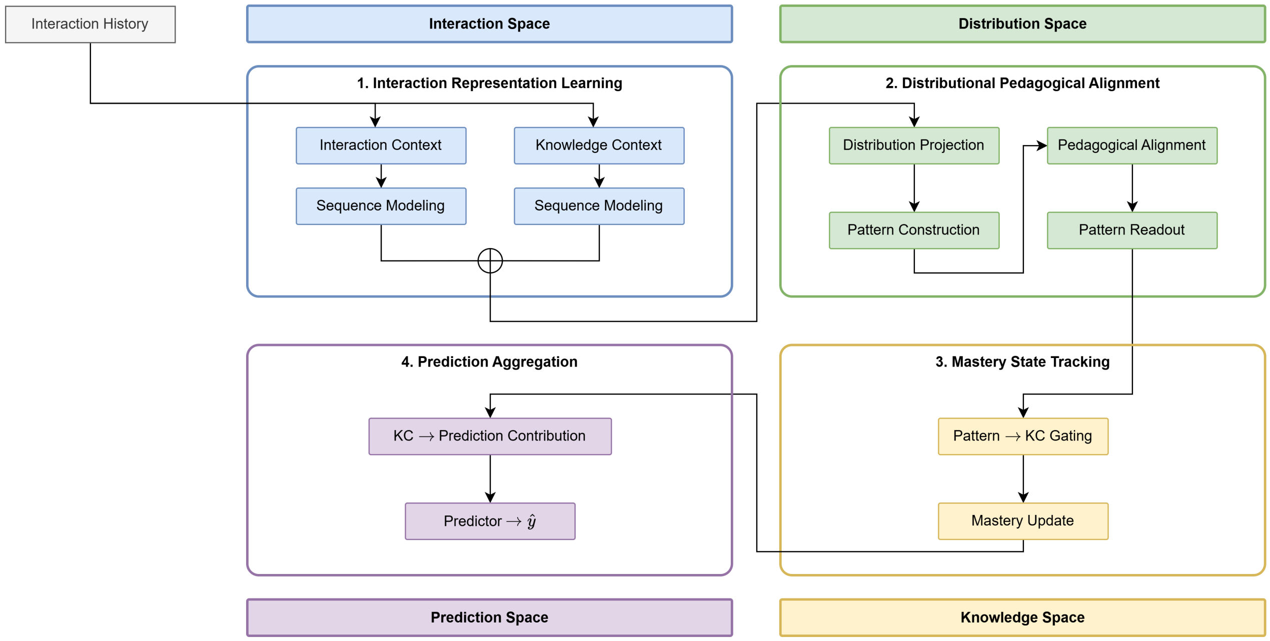

Fig. 1. Framework overview. Four modules over four representation spaces (color-coded). Evidence is *produced* in Module 2, the mastery state is *updated* in Module 3, and the prediction is formed in Module 4. Modules 3 and 4 additionally expose the pattern-to-KC contribution and the KC-to-prediction contribution, while Module 2 provides the learning patterns. The mastery memory is a per-learner recurrent state $M \in \mathbb{R}^{C\times d}$ over the C knowledge components. At step t, Module 1 reads M_{t-1} and Module 3 writes M_t.

the previous step's mastery memory M_{t-1} in the context of the current question via the knowledge graph. The two groups are processed by two parallel sequence-modeling branches: one capturing the dynamics of the interaction sequence, the other the dynamics of the local knowledge state. The two representations are fused into a single interaction representation.

D. Module 2: Distributional Pedagogical Alignment

The alignment layer, DPA, proceeds in four conceptual steps:

1) *Distribution projection* maps the interaction representation to a distributional one via a statistical-descriptor decoder. The decoder is a learnable map from the interaction representation to the statistical descriptors that define a distribution, such as its mean and variance. It carries the location and spread implicit in the interaction representation into an explicit distribution.
2) *Pattern construction* builds several learning patterns using fixed pattern operators, so that each operator always carries the same pedagogical meaning and pattern structure is consistent across learners regardless of history length or distribution. A planned list of operators is: (i) *temporal pooling* pools the most recent interactions; (ii) *same-KC pooling* pools all interactions on the same KC; and (iii) *$\mathcal{G}$-aware pooling* pools interactions following the KC topology.
3) *Pedagogical alignment* aligns the patterns with pedagogical principles. Crucially, this alignment does not directly infer mastery. Instead, each principle is realized as a *rule-shaped* transform of the pattern's distributional descriptor: the functional form of the transform is fixed by the principle. It encodes the pedagogical constraint, while only its adjustment amplitude is learnable. A planned list of principles is: (i) *difficulty-response consistency* reweights each interaction's contribution to a pooled pattern by its difficulty, grounding the adjustment in item response theory; (ii) *KC-relation consistency* adjusts the overall pattern distribution according to relations between KCs read from $\mathcal{G}$ (e.g., prerequisite); (iii) *guessing and slipping*: when a pattern of the same KC and similar difficulty contains an outlier response (a correct answer amid errors, or vice versa), the outlier is treated as a possible sign of guessing or slipping and the pattern is adjusted to absorb this possibility.
4) *Pattern readout* maps the aligned patterns back to a structured representation z', in which each block corresponds to one pattern type. The readout preserves the statistical descriptor (i.e., the mean and variance) to carry over into the mastery update.

E. Module 3: Mastery State Tracking

The third module maintains an explicit knowledge state and updates it from the structured pattern representation. Because the readout preserves the statistical descriptor, the update is uncertainty-weighted: evidence with higher variance yields a smaller mastery increment. Alongside the update, it learns a *pattern-to-KC gating* that quantifies how much each learning pattern contributes to each related KC, and the state advances by a residual increment.

F. Module 4: Prediction Aggregation

The final module predicts the outcome of the next interaction from the updated knowledge state and a representation of the target question. Alongside the probability, it produces a *KC contribution* (contribution of each KC to the target interaction) and forms the final prediction.

V. Preliminary Evidence

We report a preliminary evaluation of the framework's initial implementation. The implementation includes all four framework modules as presented in Section IV, but the specific components remain simplified and have not yet been individually validated or optimized. Consequently, the results should be interpreted as an initial feasibility study rather than evidence of the framework's final performance.

For this preliminary study, we evaluate the framework on four representative datasets selected from the two dataset groups: two flat-tag benchmarks requiring a constructed concept graph (ASSISTments 2009 [15] and Algebra [16]), and two benchmarks providing a native concept structure (XES3G5M [17] and Eedi [18]). We compare against the foundational BKT [1] and two widely adopted deep knowledge tracing baselines, DKT [2] and DKVMN [19]. More recent state-of-the-art methods are reserved for the comprehensive evaluation described in Section VI. The framework results are reported in Table I as the mean $\pm$ standard deviation over five-fold cross-validation after 200 training epochs.

The preliminary results reveal a consistent difference between the two dataset groups. On the flat-tag benchmarks, where the concept graph must first be constructed, the current implementation remains below DKT and DKVMN by approximately 2.9% to 5.5% in relative AUC. In contrast, on datasets providing native concept structures, the framework achieves the best performance, outperforming both deep baselines by approximately 2.7% to 3.5% in relative AUC. Although these results are preliminary, they are consistent with the central motivation of the proposal: explicitly modeling structured concept relationships appears to provide greater benefit when reliable concept structures are available. A comprehensive evaluation against recent state-of-the-art KT methods, together with ablation studies and statistical significance analyses, is left to the planned evaluation presented in Section VI.

VI. Planned Evaluation

Datasets. We plan to work in the mathematics domain, where the concept structure the pedagogical operators rely on is best established. Datasets are organized by whether that structure is available natively. Some benchmarks ship a genuine graph or tree, including Junyi Academy [21], XES3G5M [17], and Eedi/NIPS34 [18]. Others are widely used flat-tag benchmarks, i.e., ASSISTments [15] and Algebra/Bridge-to-Algebra [16], which we retain for scale and comparability. For the flat-tag benchmarks, we construct a graph following the validated procedure of CMDKT [4] and treat it as a fixed input (Section III).

Baselines. We compare against the foundational BKT [1] and against representatives of the three themes of Section II. These span classic and attention-based deep KT (DKT [2], DKVMN [19], AKT [22]), graph- or representation-enriched models (GKT [8], CMDKT [4]), distributional models (UKT [5], KeenKT [6], S^{2}KT [7]), reasoning or pattern-based models (PSI-KT [9], NSKT [10], PLKT [11]), and learning-process models with explicit forgetting (LPKT [14], MemoryKT [23]).

Metrics and Ablations. We report the standard AUC, accuracy, and RMSE. We add expected calibration error (ECE) and negative log-likelihood (NLL) to probe the distributional components. We plan the following ablations: (i) replacing the distributional representation with a point vector; (ii) removing the alignment step; (iii) enabling the pattern operators one at a time; and (iv) enabling the alignment principles one at a time (difficulty-response consistency, KC-relation consistency, and guess/slip). These probe the distributional space, the pedagogical alignment, and the pedagogically structured patterns, and correspond to the elements of the precondition chain of Section IV-A that can be removed independently.

Hypotheses. We expect four outcomes. (H1) The framework is competitive with strong baselines and improves most clearly where a reliable concept structure lets the pedagogical operators contribute. (H2) The relative gains are largest in data-scarce regimes, where learning patterns and pedagogical priors act as regularizers; we construct scarcity both by subsampling interactions and by a cold-start protocol that retains only the first k interactions per learner. (H3) The pedagogical modules contribute positively in ablation, and the alignment adjustment shows behavioral consistency with the pedagogical rules. (H4) Performance degrades gracefully as the concept structure is made noisier, which we induce with edge-level graph perturbations (edge rewiring and edge addition/deletion).

Potential for Interpretability. The pattern-to-KC contribution and KC-to-prediction contribution can also be chained into a model-level attribution trace read from the model's own forward pass. We treat this only as an optional direction, not a contribution of the present design. Attention- or gating-style weights are not guaranteed to be explanations of the internal mechanism [24], [25], so we frame any such reading as behavioral consistency with pedagogical rules rather than a claim about the model's internals. A dedicated evaluation is left to future work.

VII. Conclusion

We have presented a perspective on knowledge tracing in which updating knowledge is the interpretation of evidence, realized through an explicit intermediate layer, DPA, that pools interactions into learning patterns whose structure is consistent across learners and aligns them with pedagogical principles prior to mastery update. The framework's central addition is exactly DPA: the pedagogical alignment of learning patterns. As an ongoing research effort, our immediate research goals are to implement the framework and evaluate it on

TABLE I
PRELIMINARY AUC COMPARISON ON FOUR REPRESENTATIVE DATASETS.

	Flat-tag		Native structure	
Method	ASSISTments 2009 [15]	Algebra [16]	XES3G5M [17]	Eedi [18]
BKT [1]	0.6700	–	–	–
DKT [2]	**0.7541**	**0.8149**	<u>0.7852</u>	<u>0.7689</u>
DKVMN [19]	<u>0.7473</u>	<u>0.8054</u>	0.7792	0.7673
Ours	0.7128 ± 0.0080	0.7818 ± 0.0063	$\mathbf{0.8065 \pm 0.0027}$	$\mathbf{0.7912 \pm 0.0003}$

The best result in each column is shown in **bold**, and the second best is <u>underlined</u>. Established results for BKT, DKT, and DKVMN are taken from their pyKT implementation [20], while our proposal is reported as the mean ± standard deviation over five-fold cross-validation. All training and evaluation paradigms are consistent.

the benchmarks outlined in Section VI; to study mechanisms that encourage the intended role separation; and to relax the knowledge graph toward a learnable topology.

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